

TECHNICAL — DATA SCIENCE

Flight Tracking Analysis

Exploratory and predictive analysis of 10,000 U.S. commercial flights in 2024 · R (ggplot2, gbm, PCA, linear regression)

10,000	17%	6.3 min	3
FLIGHT-LEVEL OBSERVATIONS	PEAK ROUTE CANCELLATION RATE (TPA→ATL)	EXTRA PEAK-SEASON DELAY VS. SHOULDER SEASON	ANALYSIS LENSES: CARRIER, TIME, GEOGRAPHY

Background & Objectives

Flight delays and cancellations are a persistent challenge in aviation, and 2025 brought unusual shocks — a record federal shutdown that strained air traffic control staffing, alongside high-profile safety incidents that shook public confidence in air travel. This project analyzes a 2024 sample of U.S. commercial flights to surface delay and cancellation patterns across airlines, airports, routes, time, and geography, then builds models to predict whether — and how long — a flight will be delayed.

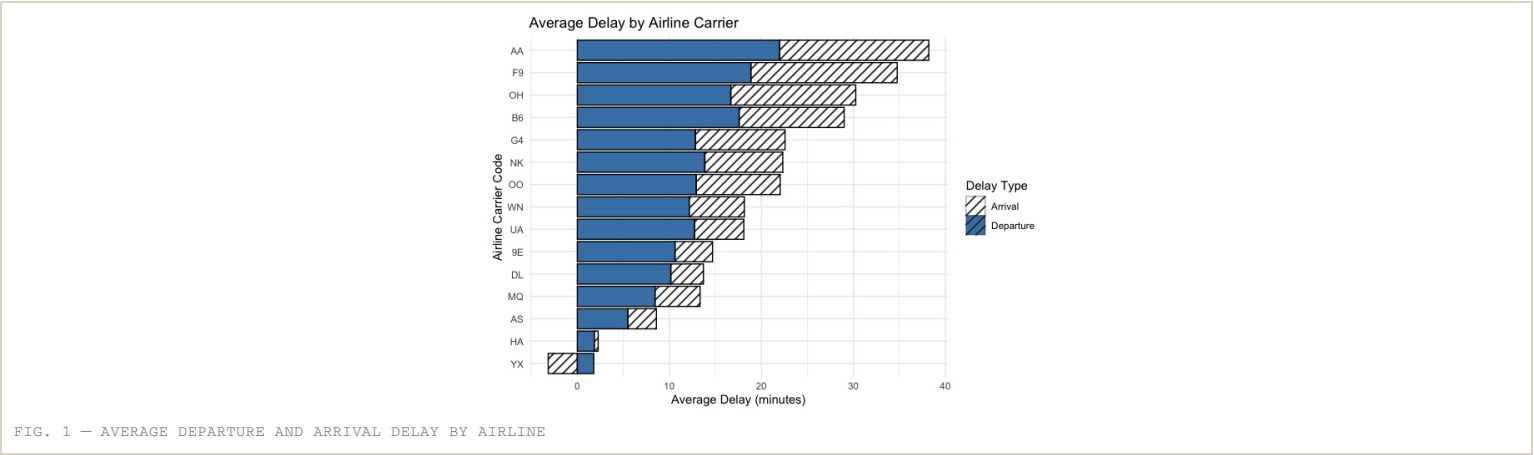
The predictive models achieved acceptable overall accuracy but poor specificity, largely driven by class imbalance (far more on-time than delayed flights). This led to a two-track approach: unsupervised learning to diagnose *why* delay occurrence is hard to predict, and linear regression to explain delay *length* once a delay occurs — where travel season and departure hour emerged as strong, significant predictors.

Methodology

The core dataset is a 10,000-row sample of 2024 U.S. commercial flights (airline, airport, route, delay times, cancellation status, and delay-cause categories), sourced from Kaggle. It's paired with an airport geolocation dataset for mapping, and monthly domestic passenger volume data (U.S. Bureau of Transportation Statistics, 2015–2025) used to build seasonal travel buckets and identify pre-/post-COVID differences.

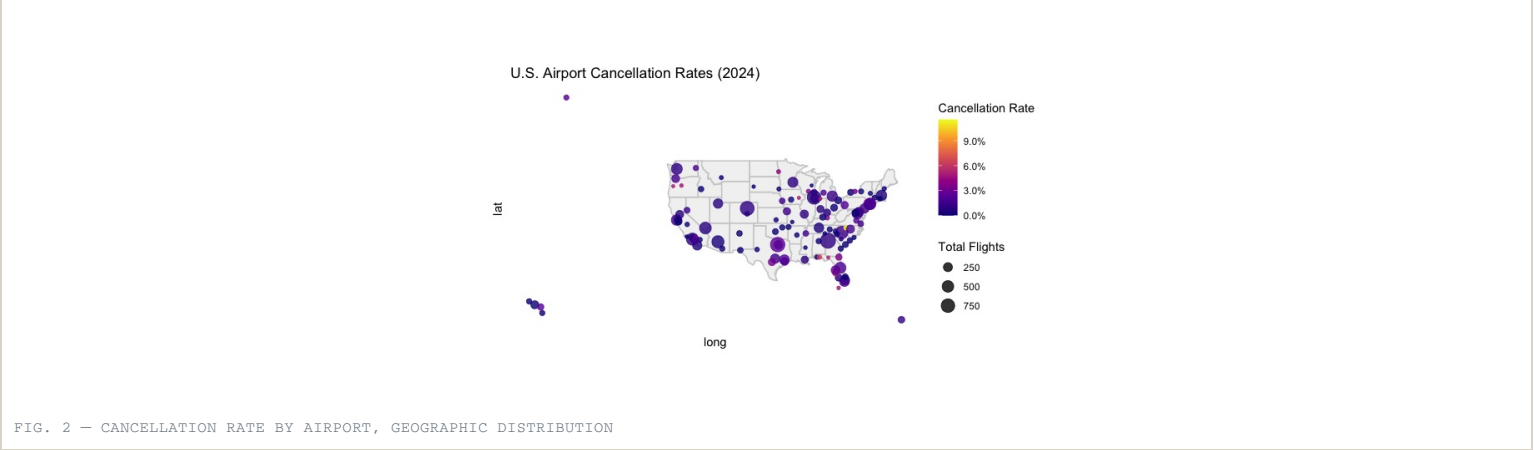
I - III. Delay & Cancellation Patterns

By carrier & route. American Airlines showed the highest cumulative delay (~40 min), with Frontier and PSA close behind; Republic Airways was a notable outlier, consistently arriving ahead of schedule. At the route level, Newark→Chicago had the worst delays (~90 min departure / 75+ min arrival), while Tampa→Atlanta had the highest cancellation probability at nearly 17% — almost double the next-closest routes.



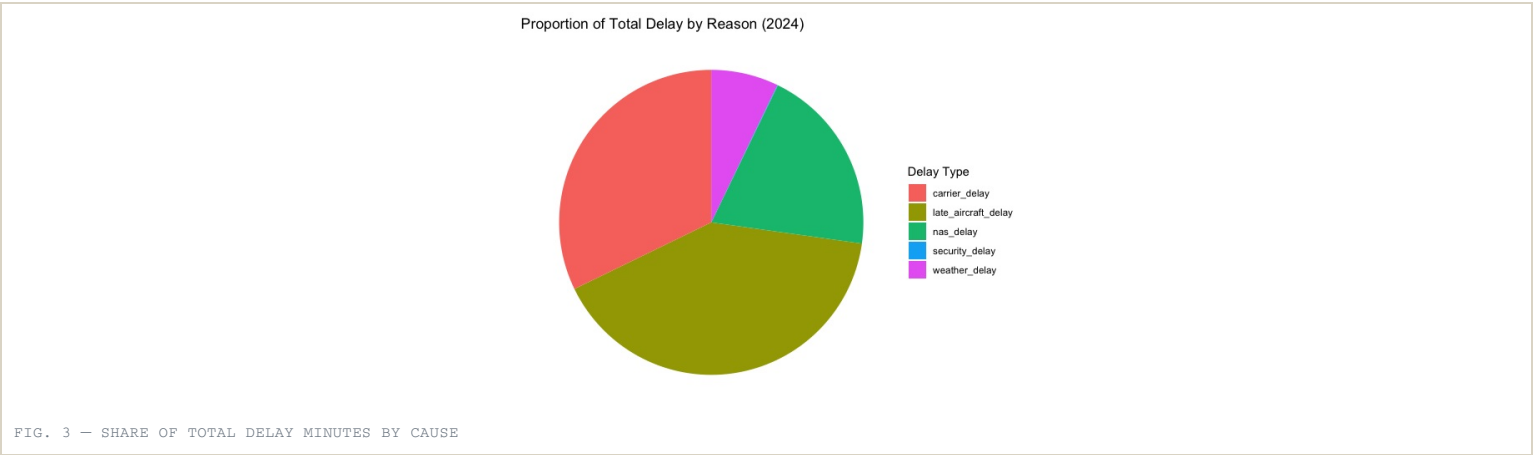
By time. Tuesday and Wednesday were the most reliable days (~10 min average delay); Saturday was the worst (16+ min) despite not being the highest-volume day. Delays were lowest 5–11 AM and climbed through the day, peaking near 8 PM; cancellations spiked overnight, exceeding 3% around 10 PM as crews hit work-hour limits and rebooking options narrowed.

By geography. Cancellation risk was concentrated regionally — hotspots near 9% in the Northeast and parts of the Southeast, versus near 0% on the West Coast. Arrival delays showed a clear East/West split (15–25 min in the Northeast corridor and Florida vs. 0–10 min in the West), driven by traffic density and airspace congestion. Dallas-Fort Worth led both total cancellations and total delayed flights, well above other major hubs.



IV. Reasons for Delays & Cancellations

Contrary to the common assumption that weather drives most delays, **late aircraft** (the ripple effect of an earlier delay) was the largest single cause of delay minutes, followed by carrier-controlled issues like maintenance and crew scheduling — together outweighing weather and National Air System delays combined. This cascading pattern matched what showed up at the hub level: late-aircraft delays dominated at major connectors (ATL, DEN, DFW), while carrier-specific issues stood out at LAX.



V. Predicting Whether a Flight Is Delayed

A gradient boosting model was trained to predict a 15+ minute departure delay using airline, day of week, departure hour, travel season, and a day-of-week $\times$ time-of-day interaction term. The model achieved high overall accuracy but very limited specificity — it defaulted to predicting "on-time" for nearly every flight, a direct consequence of class imbalance rather than a modeling flaw.

To diagnose this, a Principal Component Analysis was run on the same features. Delayed and on-time flights showed substantial overlap across all principal components, indicating the classes simply aren't well separated by scheduling variables alone — delays appear to be driven more by stochastic operational factors (weather, aircraft availability, ATC decisions) that this dataset doesn't capture.

VI. Predicting Delay Length

A linear regression on departure delay length (minutes) used airline, day-of-week, departure hour, and season as predictors. The model was jointly significant ($F = 8.10$, $p < 0.001$) but explained only 2.3% of variance in delay length ($R^2 = 0.023$) — expected, given how much of delay severity depends on unobserved factors like weather and ATC congestion.

- **Departure hour:** each later hour added ~ 1.01 minutes of delay on average, consistent with delays compounding across the day.
- **Peak season (Mar-Aug):** 6.27 minutes longer than shoulder season ($p < 0.001$).
- **Day of week:** weak effects overall; departure time and season mattered more than which day a flight was scheduled.

Out-of-sample performance: RMSE of 42.3 minutes, MAE of 23.0 minutes — useful for describing structural patterns, not for precise single-flight predictions.

Conclusion & Next Steps

Delays and cancellations concentrate at major connecting hubs (DFW, ATL, DEN), skew later in the day and toward weekends, and cluster in the Eastern U.S. Operational cascade effects (late aircraft, carrier issues) outweigh weather as a root cause. As a pilot-scale study, the clearest next step is incorporating weather and airport-congestion data, which would likely improve both delay-occurrence classification and delay-length precision.